

BAVARIA HIGHWAY TRAFFIC FORECASTING

Forecasting Alpine highway traffic, a year ahead

An automated, data-driven replacement for the hand-built Fahrkalender — daily congestion forecasts for Bavaria's two busiest Alpine motorway corridors.



THE PROBLEM

Replacing a calendar built by hand

Every year, Die Autobahn experts hand-build the **Fahrkalender** — a colour-coded calendar of expected holiday traffic. It is manual, slow to update, and published only months ahead.

Our system learns the same judgement from three years of detector data, then issues it automatically: every day, every corridor, up to a year in advance.

85,094

vehicles on the single busiest day recorded

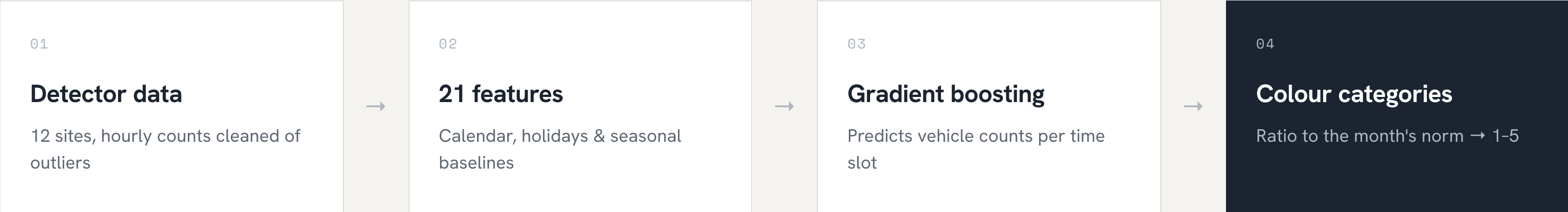
283,160

hourly detector readings, 2023–2025

1 year

forecast horizon, from the calendar alone

How the forecast is built



■ **Judged against the month, not the year**

A busy July day is measured against typical Julys — so a quiet midnight never reads like afternoon rush hour.

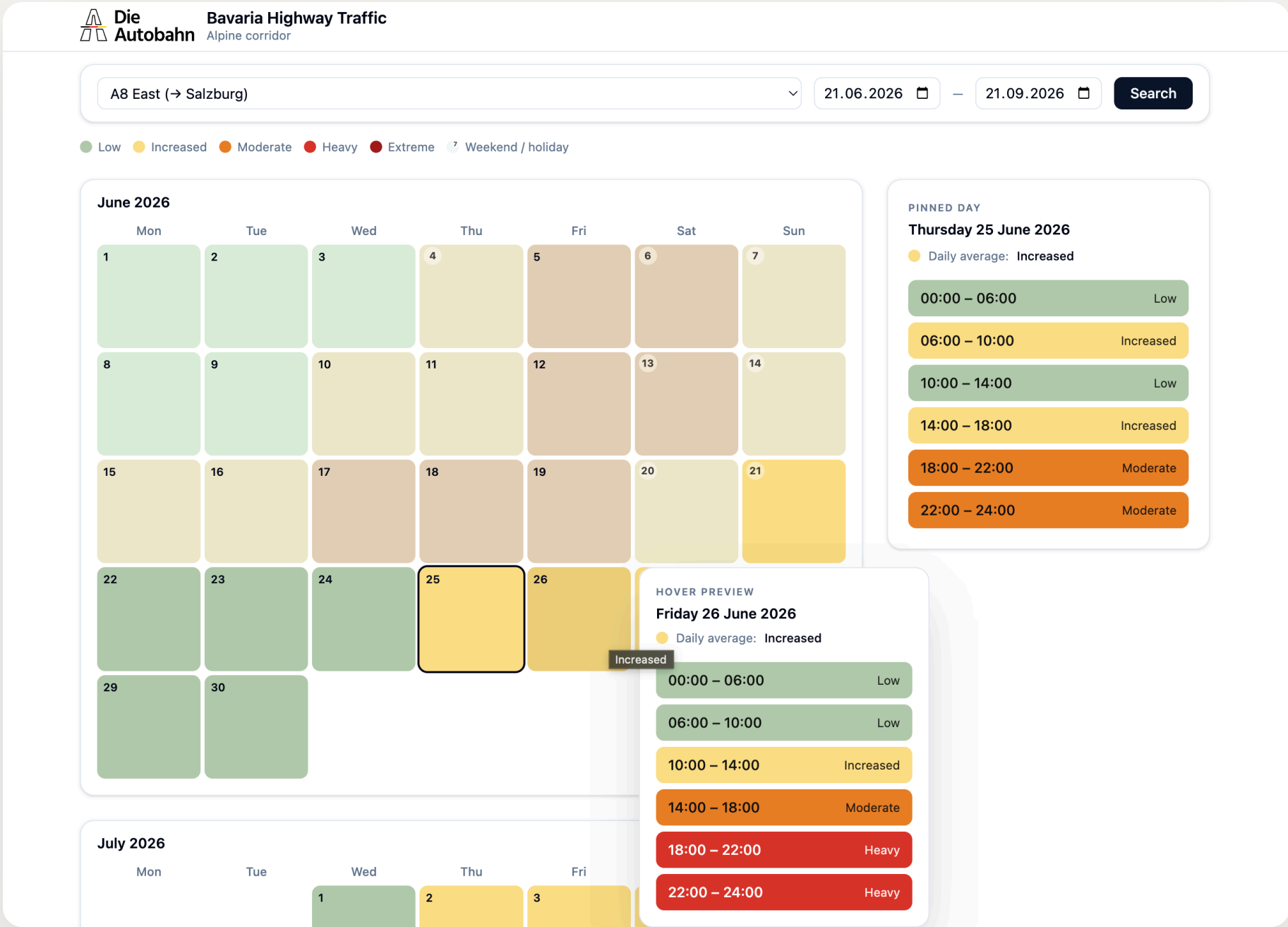
■ **Tested on a year it never saw**

Trained on 2023-24 and validated on a fully held-out 2025 — no peeking at the future during training.

What every traveller sees

	Low	Free-flowing
	Increased	Slightly above normal
	Moderate	Congestion risk
	Heavy	Expect delays
	Extreme	Significant congestion

Pick a road, direction and date range. Each day takes the average of its six time slots — with a slot-by-slot breakdown and the reasons behind it on click.



THE RESULTS

How well it predicts

0.923

R^2 — variance explained on held-out 2025

558

mean absolute error, vehicles per
time slot

13.0%

mean absolute percentage error
across that year

8,233

held-out slot-days, none of them seen
in training

Trained on 2023–24 and scored on 2025 — the same task the system faces
in production: forecasting a year it has never seen.